

G. NARAYANAMMA INSTITUTE OF TECHNOLOGY & SCIENCE
(Autonomous) (For Women)
Department of Computer Science and Engineering
IV B.Tech I Semester
Data Science Using R Lab
Academic Year 2026 – 2027

Course code: 12753

Date: 6-07-2026

Lab Cycle-1

1) Calculator Application:

- a) Using with and without R objects on console.
- b) Using mathematical functions on console.
- c) Write an R script, to create R objects for calculator application and save in a specified location in disk.
- d) R Packages: tidy, ggplot2, ggraph, glue. Shiny.

2) Descriptive Statistics:

- a) Write an R script to find basic descriptive statistics using summary.
- b) Write an R script to find subset of dataset by using subset.

3) Reading and Writing Different Types of Datasets:

- a) Reading different types of data sets (.txt, .csv) from web and disk and writing in file in specific disk location.
- b) Reading Excel data sheet in R.
- c) Reading XML dataset in R.

4) Visualizations:

- a) Find the data distributions using box and scatter plot.
- b) Find the outliers using plot.
- c) Plot the histogram, bar chart and pie chart on sample data

5) Correlation and Covariance:

- a) Develop a program to find the correlation matrix on iris data.
- b) Plot the correlation plot on dataset and visualize giving an overview of relationships among data on iris data
- c) Analysis of covariance: variance (ANOVA), if data has categorical variables on iris data.

6) Write a R-script that detects file type (.csv, .txt, .xlsx, .xml) and reads it accordingly.

Note: The programs which are mentioned in bold case are extra programs

Staff Incharges

Dr. D V Lalitha Parameswari, Associate Professor
Dr. G Malini Devi, Associate Professor
Dr. T Rajesh, Assistant Professor
Dr. N Anil Kumar, Assistant Professor

HOD CSE

Dr. A Sharada

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Lab Cycle-2

1) Regression Model:

- a) Import a data from web storage. Name the dataset and now do Logistic Regression to find out relation between variables that are affecting the admission of a student in an institute based on his or her GRE score, GPA obtained and rank of the student. Also check the model is fit or not. Require (foreign), require (MASS).
- b) Apply regression Model to predict the data on any dataset.

2) Multiple Regression Model:

- a) Develop Multiple Regression model for any dataset.
- b) Apply Multiple Regression Model to predict the data on any dataset.

3) Classification Model:

- a) Install relevant package for classification.
- b) Implement and evaluate the performance of various classifiers with different datasets.

4) Clustering Model:

- a) Clustering algorithms for unsupervised classification.
- b) Plot the cluster data using R visualizations.

5) Time Series Model:

- a) Plot the Time series data using R visualizations.
- b) Time series Forecasting using Exponential smoothing, ARIMA models.

6) Perform a complete data analysis on the built-in airquality dataset, including import, cleaning, and visualization.

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